L.no = 5

Floatation.

→ **Fluids-** It is a substance which flow under the action of Applied force. (Liquid + Gas).

→ **Pressure-** Normal force acting per unit area on the surface of body.

$$P = \frac{F}{A} \rightarrow \frac{\text{Trust}}{A}$$

⇒ ~~Thrust~~ Trust = Normal force on the surface of body.

⇒ Unit of Pressure- $P = \frac{F}{A} = \frac{N}{m^2} \Rightarrow \boxed{\text{Pascal}}$

⇒ **Density-** ρ (Rho)

$\rho = \frac{m}{V}$ - density is defined as mass per unit volume.

- Unit- kg/m^3

☆⇒ Density of water = $1000 kg/m^3$ at $4°C$.

☆→ ~~D.Rela~~ Relative density: $\frac{\text{Density of the substance}}{\text{Density of water}}$

- Density of substance = R.D × Density of water.

Q) R.D of gold is 19.6

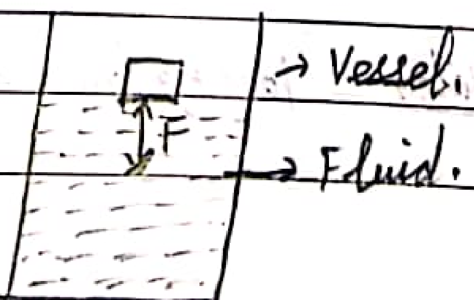
∴ Density of gold = R.D × Density of substance
= $19.6 \times 1000\ kg/m^3$
= $19600\ kg/m^3$.

⇒ Gold is 19.6 times hevier then equal volume of Water.

Q- R.Density of silver (Ag) is 10.3.

- Density of silver = R.D × D. of Water
= $10.3 \times 1000\ kg\,m^{-3}$
= $10300\ kg\,m^{-3}$

⇒ Buoyancy and Buoyant force:-



- The upward force exerted by a liquid on the surface of object, when it is immersed by a fluid is called buoyant fore/Upward Thrust.

⇒ <u>Archimede's principle</u>- When a body is immersed partially or fully it experiences a upward force which is equal to the weight of the displaced fluid by the immersed portion of the body.

$$W = mg$$
$$\downarrow$$
$$T = W = mg$$
$$\boxed{T = W = \rho V g}$$

Sound

⇒ **Sound** - Sound is a wave produced by the viberating object.

- It is a longitudinal wave.

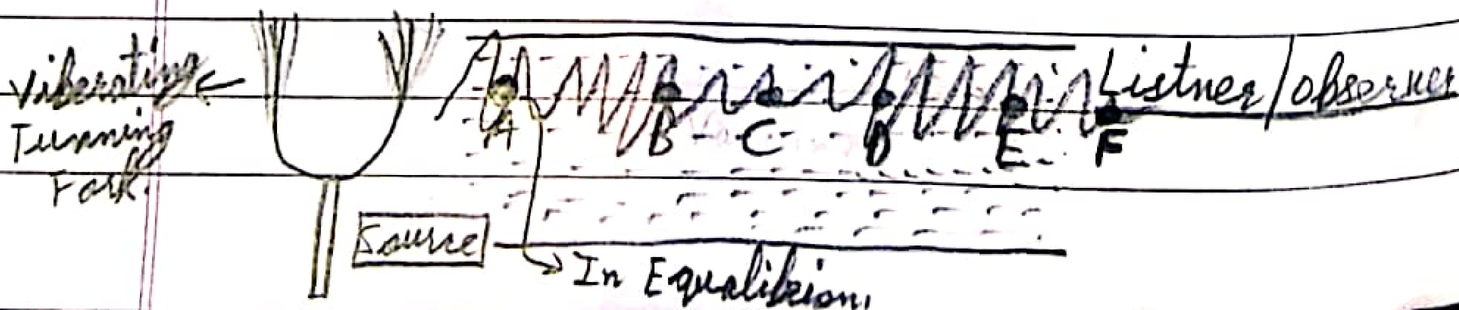
⇒ **Production of sound** - Sound is produced by the viberating object.

e.g Viberating tunning fork.

⇒ **Propagation of Sound** - The sound needs medium (matter or surrounding) for its propagation i.e Solid, Liquid, gas.

⇒ Sound can not propagate through Vacuum.

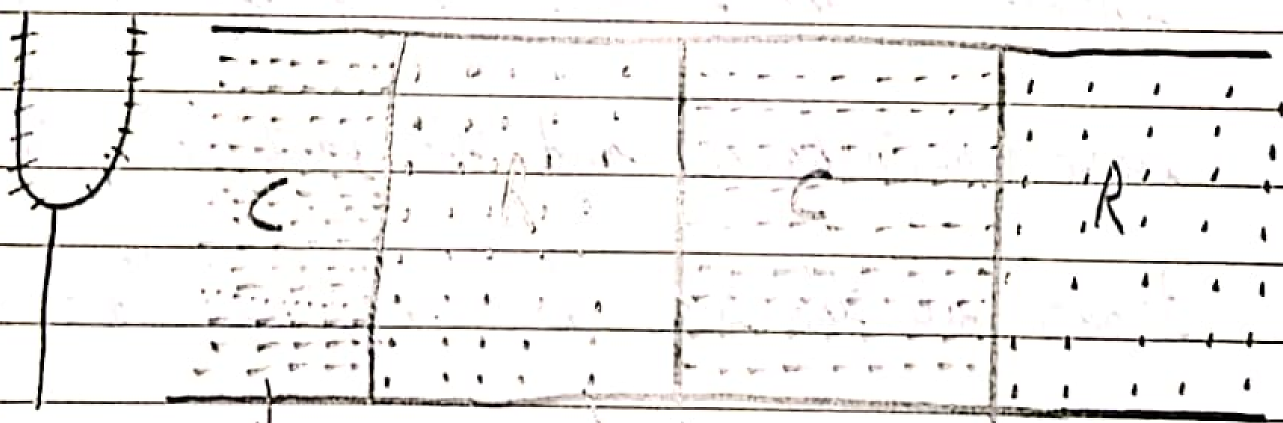
⇒ How it Propagates -



- The particles are viberating with greater Amplitude therefore disturbs the equalibrium of the neighbour particle.

Note:- The particle did not move/travel all the way from source to the ~~bsb~~ listener but the disturbance travel from source to listener.

⇒ **Sound is a longitudinal:-**



Compression. Rarefaction.

→ Compression:- The Region of high pressure in a longitudinal wave is called as compression/

The Region where density of particles is high.

→ <u>Rarefaction</u>:- The region of low pressure in a longitudinal wave is called rarefaction. OR the region where density of particles of medium is low.

→ In Longitudinal wave the particles of motion vibrates in the direction of propagation of wave (disturbance) and the particles did not travel all the way but they oscillate about their mean position.